TEMPERATURE MEASUREMENT WITH A SONIC ANEMOMETER AND ITS APPLICATION TO HEAT AND MOISTURE FLUXES

P. SCHOTANUS

Institute for Meteorology and Oceanography, University of Utrecht, The Netherlands

F.T.M. NIEUWSTADT*, H.A.R. DE BRUIN

Royal Netherlands Meteorological Institute, De Bilt, The Netherlands

(Received in final form 14 March, 1983)

Abstract. The possibility of measuring heat and moisture fluxes using sonic anemometer data is investigated. Theoretical relations for the temperature variance and heat flux are derived. In the first part of this paper, these relations are verified by experimental data, involving a sonic anemometer, a fast thermocouple and a Lyman-α hygrometer. In the second part we propose two simple procedures to estimate heat flux from sonic anemometer data. The first one requires a rough estimate of the Bowen ratio; for the second one the net radiation is needed. Using the last method, a good estimate of the moisture flux is also obtained.

1. Introduction

The measurement of the turbulent fluxes of momentum, heat and moisture is an important objective in experimental boundary-layer research. For these measurements, the sonic anemometer has become an important research tool (Kaimal, 1979). Sonic anemometers have been used in many experiments for the measurement of wind velocity fluctuations (Haugen *et al.*, 1971; Kaimal *et al.*, 1976); modern sonic anemometers also provide temperature measurements, but these are seldom used for flux computations (Kaimal, 1969; Friehe, 1976).

In this paper we explore the merits of the sonic anemometer for the measurement of temperature fluctuations. For that reason we conducted a field experiment in the summer of 1981 near the meteorological mast at Cabauw, the Netherlands. Apart from a sonic anemometer, we employed a fast thermocouple and a Lyman-α hygrometer. With this array of instruments, we could measure the three vertical turbulent fluxes: momentum, heat and moisture.

The first problem that we were faced with, involved errors in the velocity measurements due to flow distortion by the sonic anemometer. These errors are caused by the wake of the sound transducers, and result in an underestimation of the wind velocity (Kaimal, 1979; Wyngaard, 1981). They are usually neglected. Here, we have used a calibration curve to correct for this problem.

Subsequently, we investigated the use of a sonic anemometer to measure temperature variance and heat flux. This possibility has not been explored much until now. However,

* To whom correspondence should be directed.

Boundary-Layer Meteorology **26** (1983) 81–93. 0006–8314/83/0261–0081$01.95.
Copyright © 1983 by D. Reidel Publishing Company.

the theoretical background has been well established (Kaimal and Businger, 1963). They showed that the sonic-derived temperature is affected by both humidity and velocity fluctuations. A first estimate of these error terms has been given by Kaimal (1969). Here, we will focus on direct observations of all correction terms. As a next step, we will show that the temperature output of a sonic anemometer can be used to compute the heat and the moisture flux.

The various aspects of this experiment will be discussed in subsequent sections: basic equations for a sonic anemometer (Section 2); instruments used in our experiment (Section 3); analysis of the experimental data (Section 4); and methods to determine heat and moisture flux from sonic anemometer data (Section 5).

2. Theory of Temperature Measurements with a Sonic Anemometer

The operating principles of a sonic anemometer are well known (Kaimal and Businger, 1963). A brief review of the equations involved in temperature measurements is presented here.

Sound pulses are transmitted back and forth between transducers (Figure 1). Assuming a uniform wind and temperature field, the transit times for the two pulses are:

$$t_1 = \frac{l}{c\cos\alpha + v_l}, \qquad t_2 = \frac{l}{c\cos\alpha - v_l}, \tag{1}$$

where l is the path length, v_l is the wind component along the sound path, c is the sound velocity and $\alpha = \sin^{-1}(v_n/c)$. v_n is the wind component normal to the sound path.

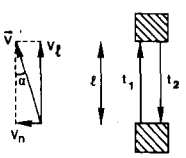


Fig. 1. Schematic drawing of a single axis sonic anemometer.

The sum of the reciprocals of the transit times is proportional to the sound velocity:

$$1/t_1 + 1/t_2 = (2c\cos\alpha)/l. \tag{2}$$

The sound velocity is given by

$$c^2 = \gamma RT(1 + 0.51q), \tag{3}$$

where T is the temperature and q is the specific humidity. For air, γR equals 403 $m^2\,s^{-2}\,K^{-1}$. This equation shows that the sound velocity is temperature dependent. Therefore, the sum of the reciprocals of the transit times (2) can be used to estimate temperature.

The sonic anemometer temperature output, T_s, is then obtained from the following equation

$$T_s = \frac{l^2}{4\gamma R}\left\{\frac{1}{t_1} + \frac{1}{t_2}\right\}^2. \tag{4}$$

Next we separate the terms in (2), (3), and (4) into their mean and fluctuating parts (denoted by overbars and primes, respectively): $c = \bar{c} + c'$, $\cos\alpha = \overline{\cos\alpha} + (\cos\alpha)'$, $T = \overline{T} + T', q = \bar{q} + q'$. Furthermore, we use a first-order Taylor expansion (remember $\cos\alpha = \{1 - v_n^2/c^2\}^{1/2}$). After substitution, the mean and fluctuating parts of T_s are

$$\overline{T}_s = \overline{T}(1 + 0.51\bar{q}) \tag{5}$$

and

$$T_s' = T' + 0.51q'\overline{T} - \frac{2\overline{T}}{c^2}\bar{v}_n v_n'. \tag{6}$$

This equation shows that the sonic anemometer only measures the real temperature fluctuation, T', when the humidity and normal velocity fluctuations are zero. Most sonic anemometers measure temperature along the vertical axis so that v_n is the horizontal wind speed u.

We now calculate the variance of T_s and the correlation of T_s with the vertical velocity w.

$$\sigma_{T_s}^2 \equiv \overline{T_s'^2} = \sigma_T^2 + 1.02\overline{T}\,\overline{q'T'} + (0.51)^2\,\overline{T}\,\sigma_q^2 + 4\frac{\overline{T}^2\overline{u}^2}{(\overline{c^2})^2}\sigma_u^2$$

$$-\frac{4\overline{T}\overline{u}}{\overline{c^2}}\overline{u'T'} - 2.04\frac{\overline{T}^2\overline{u}}{\overline{c^2}}\overline{u'q'}, \tag{7}$$

$$\overline{w'T_s'} = \overline{w'T'} + 0.51\,\overline{T}\,\overline{w'q'} - 2\frac{\overline{T}\overline{u}}{\overline{c^2}}\overline{u'w'}. \tag{8}$$

These are the basic equations, which will be verified in Section 4 with the aid of our experimental data.

3. Experimental Set-up

In the summer of 1981, a field experiment was conducted near the meteorological mast at Cabauw, the Netherlands. A full description of the surroundings of the mast and its instrumentation is given by Driedonks *et al.* (1978). For the purpose of this study, it is relevant to mention that within a radius of 200 m, the surroundings of the mast consist of flat grassland. Beyond this radius, obstacles, generally in the form of trees, are present. The instruments employed during the experiment are briefly described below.

The three-dimensional sonic anemometer-thermometer used was made by Kaijo-Denki (model DAT–300). The sensor (model TR–61) has two horizontal sound paths crossing at 120° and a vertical sound path. The length of these sound paths is 0.2 m, and the diameter of the sound transducers is about 15 mm. A more detailed description is given by Hanafusa (1980). Velocities measured by a sonic anemometer are underestimated due to the wake of the sound transducers. The results of a wind-tunnel study to estimate the magnitude of this effect are discussed in an appendix.

Temperature fluctuations were measured by means of a fast-response thermocouple with a diameter of about 200 μm. Its unventilated response time is about 0.2 s (Driedonks *et al.*, 1978). The thermocouple was mounted within the measuring volume of the sonic anemometer.

Humidity fluctuations were measured with a Lyman-α hygrometer (model BLR from Electromagnetic Research Corp.). This instrument measures the absolute humidity using the absorption of Lyman-α radiation (121.6 nm) by water vapour (Buck, 1973, 1976). The instrument was mounted on top of the sonic anemometer. Obstacle effects on the flow, within the sonic anemometer measuring volume, were found to be negligible after a wind-tunnel test. The Lyman-α hygrometer was calibrated before and after the experiment. Due to source-tube aging and deterioration of the windows by moisture in the air, the output goes down, so the instrument is not reliable for measuring the mean humidity over periods longer than a day or so. Fortunately, we found that the slope of the calibration curve did not change, so that fast fluctuations are not affected. During our experiment, the path length was 20 mm.

The whole array of turbulence instruments was mounted on a small mast at a height of 3.5 m. Since surface-layer spectra shift towards higher frequencies with decreasing height, this height is of consequence for the errors due to limited response time or finite measuring volume. Using the method described by Kaimal (1975), we find that in our case the errors in the variances and covariances of the turbulent variables do not exceed 5%.

Finally we had observations of net radiation. These were made with a Funk radiometer (Funk, 1959), installed about 100 m away from the turbulence array above similar terrain.

The turbulence data were sampled at a rate of 10 Hz with the aid of a minicomputer (HP 21 MX). From these data, variances and covariances were calculated for averaging periods of 10 min. In the calculation procedure, a linear trend was first removed from the data.

The total experiment involved observations during several days. During each day, several 10-min runs were collected. Table I gives general information on conditions during each day.

TABLE I

Conditions during the measuring days; z is the measuring height (3.5 m) and L the Obukhov length

Run number	Stability range in terms of z/L	Average wind velocity in m s^{-1}	Number of 10-min periods
141	$-0.5 < z/L < -0.2$	3	9
160	$-0.02 < z/L < 0$	7	18
161	$-0.03 < z/L < -0.01$	7	17
183_{I}	$-0.4 < z/L < -0.05$	4	14
183_{II}	$-0.4 < z/L < -0.1$	4	18

4. Analysis of the Sonic Temperature Data

In this section we discuss the verification of relations (7) and (8) by means of the observed data.

First we consider the temperature variance σ_{T_s}. A value for σ_{T_s} is directly obtained from the sonic anemometer output. This value is related by means of (7) to the data of the fast thermocouple, the Lyman-α and the sonic anemometer wind velocity. As a result of the measurements, we found that all the terms on the right-hand side of (7) are negligible except the first two. Equation (7), therefore, can be reduced to

$$\sigma_{T_s^2} \simeq \sigma_T^2 + 1.02\,\overline{T}\,\overline{q'T'}. \tag{9}$$

Its verification is shown in Figures 2 and 3. In Figure 2 we have collected the unstable cases, while in Figure 3 the nearly neutral runs are shown. The value obtained from the sonic anemometer is called σ_{T_s} (exp). The σ_{T_s} (calc) is computed from (9), first by neglecting the $\overline{q'T'}$ term and secondly by including it. The slopes given in the figures have been obtained by linear regression with a straight line through the origin. We see that for unstable conditions, Equation (9) holds very well. On the average σ_{T_s} is about 5% larger than σ_T. In neutral conditions, relation (9) is satisfied for $\sigma_{T_s} > 0.5$. For lower values, σ_{T_s} (calc) $< \sigma_{T_s}$ (exp). A tentative explanation of this result is that the finite response time of the thermocouple causes a substantial error in the measured σ_T for this

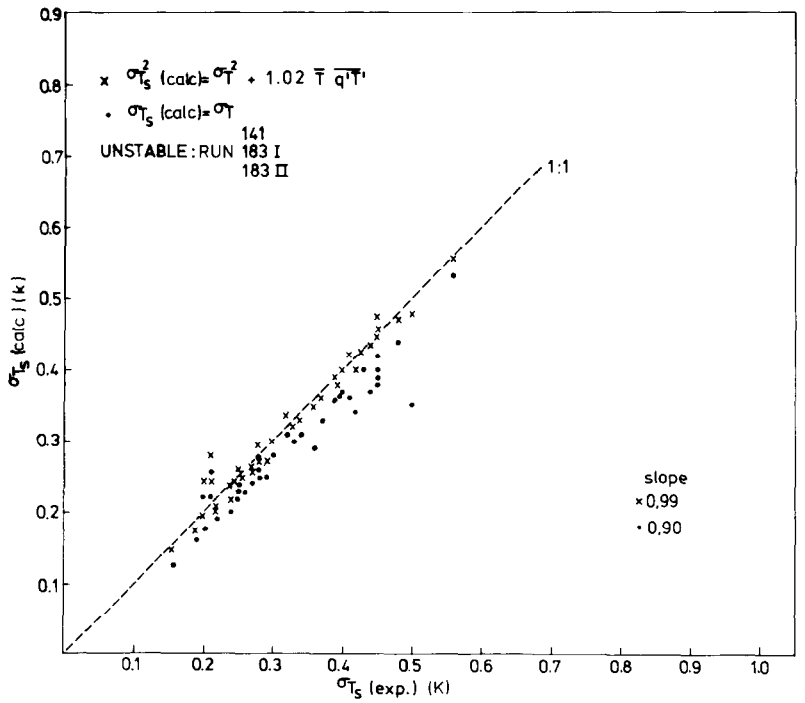


Fig. 2. The sonic temperature variance σ_{T_s} (exp) compared with the value σ_{T_s} (calc) obtained from Equation (9) for unstable conditions. The slope of a linear regression line through the origin is tabulated.

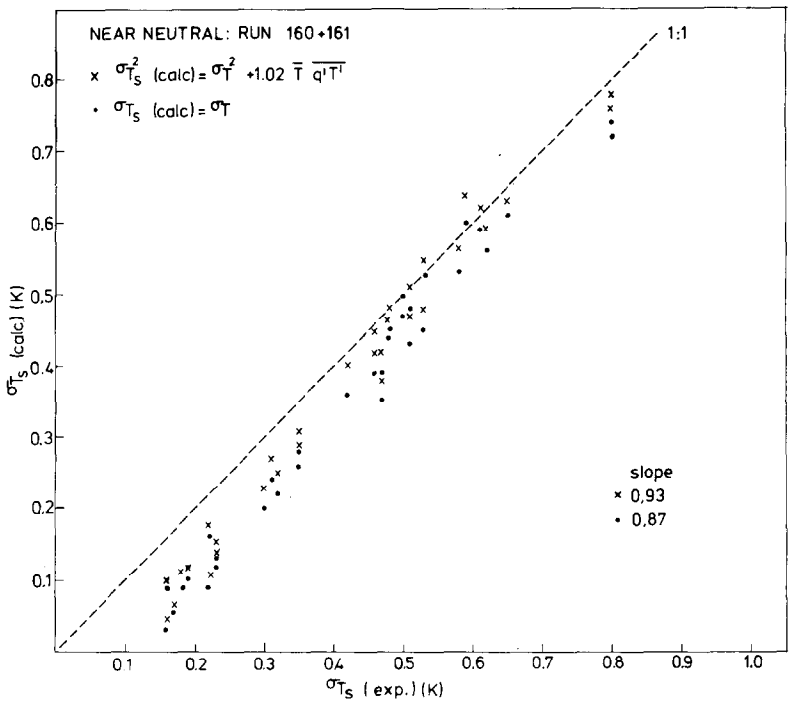


Fig. 3. The same as Figure 2 for near neutral conditions.

case. We expect a larger error in the near neutral case because compared to unstable conditions the spectral peak shifts to higher frequencies (Kaimal *et al.*, 1972).

Next we consider the covariance flux given by (8). The results are shown in Figures 4 and 5 for the near-neutral and the unstable cases, respectively.

The same meaning is attached to the notation $\overline{w'T_s'}$ (exp) and $\overline{w'T_s'}$ (calc) as introduced before. Furthermore, the influence of the separate terms in (8) is explicitly shown. The slopes in the figures have been calculated in the above described way. We find that the third term on the right-hand side of (8) is small in unstable conditions but cannot be neglected in near-neutral situations, where it amounts to a correction of about 20%. The second term in (8) has a magnitude of 10% for both cases. From Figures 4 and 5 we conclude that relationship (8) is well confirmed by the observations.

5. Estimates for the Temperature and Moisture Fluxes

Now we describe a method to determine the temperature and moisture fluxes from sonic anemometer data only. We thus assume that the fast thermocouple and the Lyman-α data are not available.

When the sonic temperature flux, $\overline{w'T_s'}$ is given, $\overline{w'T'}$ can be obtained in principle from Equation (8). The different correction terms in this equation must then be estimated. The third correction term (proportional to $\overline{u'w'}$) presents no problems,

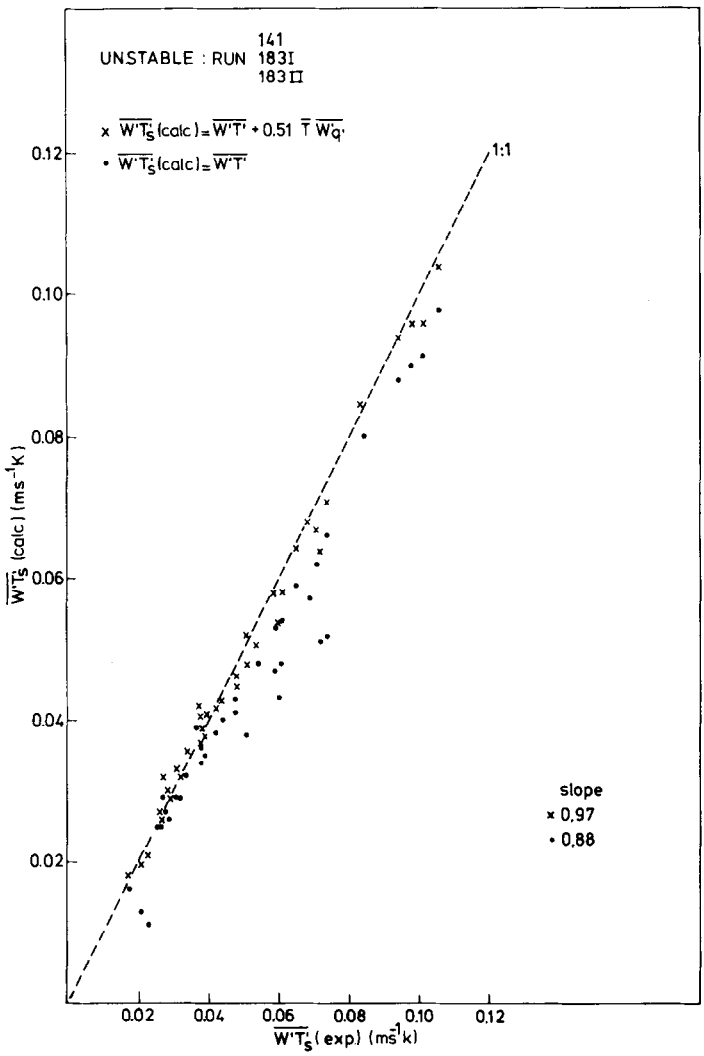


Fig. 4. The sonic heat flux $\overline{w'T'_s}$ (exp) compared with the value $\overline{w'T'_s}$ (calc) obtained from Equation (8) for unstable conditions. The slope of a linear regression line through the origin is tabulated.

because it can be evaluated directly from the sonic anemometer data itself. However, an estimate for the moisture correction term must be provided. We describe here two methods to obtain this term.

For the first method, we rewrite Equation (8) as

$$\overline{w'T'_s} = \overline{w'T'}\left(1 + \frac{0.51\overline{T}C_p}{\lambda\beta}\right) - 2\frac{\overline{T}\overline{u}}{\overline{c^2}}\overline{u'w'} \tag{10}$$

where the Bowen ratio β is given by

$$\beta = \frac{C_p}{\lambda}\overline{w'T'}/\overline{w'q'}.$$

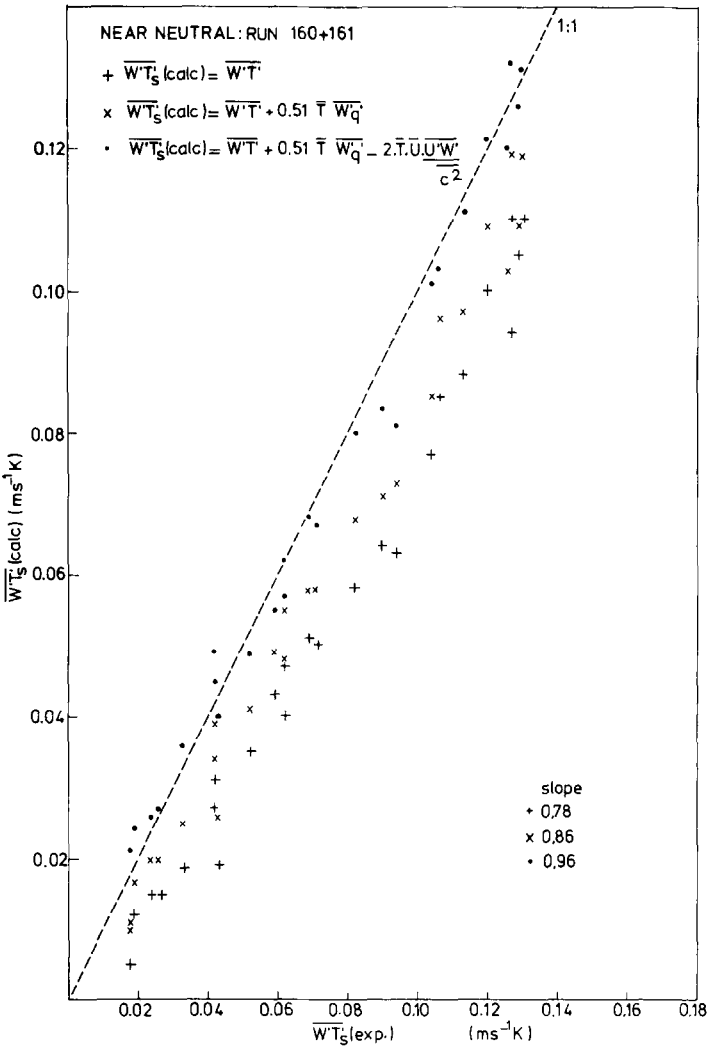


Fig. 5. The same as Figure 4 for near neutral conditions.

Here, C_p is the specific heat at constant pressure (for air ~ 1012 J kg^{-1} K^{-1}) and λ is the latent heat of condensation (for water $\sim 2.5 \times 10^6$ J kg^{-1}). We now assume a value for β. No great accuracy is required because the moisture correction term is relatively small. De Bruin and Holtslag (1982) provide a model to estimate β for a general case. For Cabauw they suggest an average value of 0.4. With this value we calculate $\overline{w'T'}$ from (10). The results are compared in Figure 6 with the measurements involving the fast thermocouple. The agreement is satisfactory.

The second method uses an additional equation between $\overline{w'q'}$ and $\overline{w'T'}$ as a supplement to (8). We choose the energy balance equation for the earth surface:

$$Q_n = \rho C_p \overline{w'T'} + \rho \lambda \overline{w'q'} + G \tag{11}$$

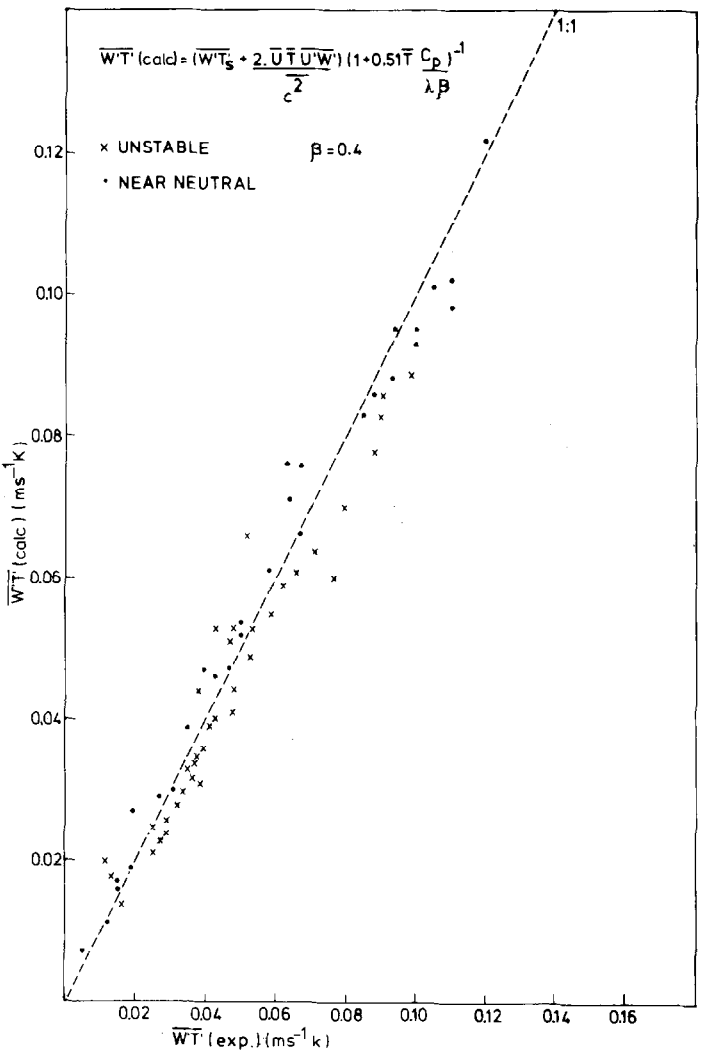


Fig. 6. The heat flux $\overline{w'T'}$ (calc) obtained from Equation (10), using a value of the Bowen ratio equal to 0.4, compared to the heat flux $\overline{w'T'}$ (exp) directly measured.

where Q_n is the net radiation and G is the soil heat flux. The air density is given by ρ ($\sim 1.2\ \mathrm{kg\ m^{-3}}$). We assume that values of Q_n are available. In our case these were obtained by means of a net radiometer. Only the value of G remains to be estimated. In general, G is small compared with Q_n during daytime conditions, to which our study is restricted. Therefore, a rough estimate of G is sufficient. De Bruin and Holtslag (1982) propose $G = 0.1\ Q_n$ for Cabauw. With this figure, we calculate the heat and moisture fluxes from the sonic heat flux $\overline{w'T_s'}$ and the net radiation. The results are compared in Figures 7 and 8, respectively, with the experimental data measured with the fast thermocouple and the Lyman-α. The agreement for the heat flux (Figure 7) is very good.

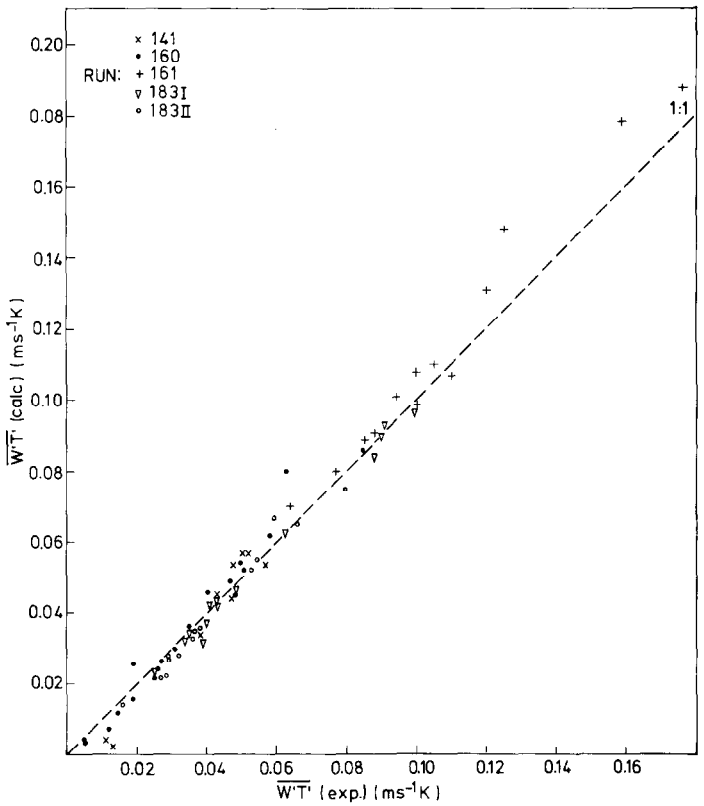


Fig. 7. The heat flux data $\overline{w'T'}$ (calc) calculated from Equation (8) and the energy balance Equation (11) ($G = 0.1\ Q_n$), compared to $\overline{w'T'}$ (exp) measured directly using a fast response thermocouple.

The moisture flux data (Figure 8) show somewhat more scatter. Furthermore, the calculated moisture flux, using (8) and (11) is slightly higher than the measurement with the Lyman-α. A reason may be that the estimate $G = 0.1\ Q_n$ is not applicable for our results (an estimate $G = 0.2\ Q_n$ removes the bias). When we take $G = 0.2\ Q_n$, this has no significant effect on the calculated heat flux since the moisture flux term gives only a small contribution in Equation (8). However, the calculated moisture flux is more sensitive to this change because it is determined as the difference between two large numbers.

Provided that acceptable values for the correction terms in (8) can be estimated, we may conclude that the sonic anemometer can be used to measure heat and moisture fluxes, a valuable extension to its normal use.

6. Conclusions

Simultaneous measurements from a sonic anemometer, a fast-response thermocouple and a Lyman-α hygrometer, were presented.

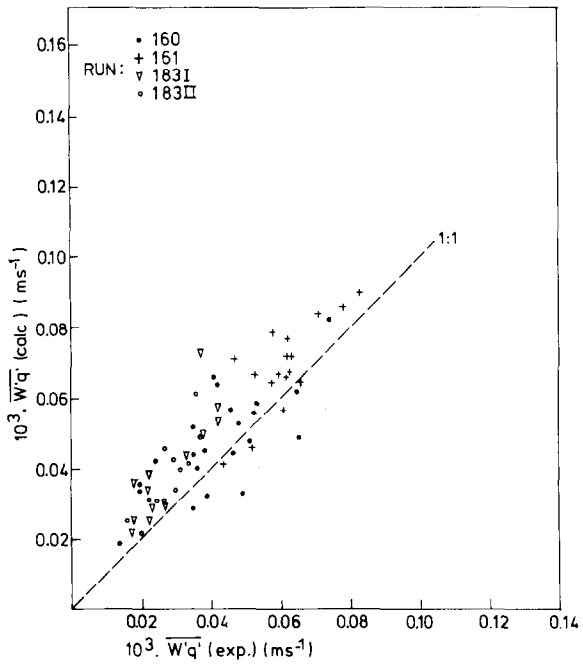


Fig. 8. The same as Figure 7 for the moisture flux: $\overline{w'q'}$ (calc) is obtained from Equations (8) and (11) and $\overline{w'q'}$ (exp) is directly measured with the Lyman-α hygrometer.

Based on a brief review of sonic anemometer theory, we derived expressions for the sonic anemometer temperature variance and heat flux. These relations were verified with the aid of direct measurements of temperature and humidity fluctuations. Excellent confirmation of the theory was found.

Finally we investigated whether the heat and moisture fluxes can be obtained from sonic anemometer data only. Two methods were proposed. The first was based on an estimate of the Bowen ratio, whereas the second employed the energy balance equation together with observations of net radiation. Acceptable estimates of heat and moisture fluxes were found. The sonic anemometer can thus also be used as a reliable instrument for measuring heat and moisture fluxes.

Acknowledgements

We are very grateful to A. C. M. Beljaars and H. Tennekes for their careful reading of the manuscript. We also want to thank W. A. Oost from the department of Oceanography for lending us the sonic anemometer. For the field experiment at Cabauw, we acknowledge the support of the Instrumental Branch of our institute.

Appendix: Calibration Results for the Sonic Anemometer.

The coordinate system of the sonic anemometer is defined in Figure A1. The velocity V is directed along the x-axis (wind-tunnel axis). The instrument measures the wind

velocity along the A and B soundpaths. The results are converted into the velocity components u_m and v_m along the x- and y-axis, respectively. For a perfect instrument we would find

$$u_m = V \cos \alpha \tag{A1}$$

$$v_m = V \sin \alpha \tag{A2}$$

where α is the angle between the X-axis and the x-axis. These relations were verified by rotating the instrument about a vertical axis. A significant deviation from the cosine relation was found for $\|\alpha\| > 10°$. For $\alpha \simeq 50°$, the error becomes in the order of 20%. The u-component calibration results are well followed by the curve

$$u_m = V\{-1.8 \times 10^{-9}\alpha^4 + 9.3 \times 10^{-8}\alpha^3 - 1.8 \times 10^{-4}\alpha^2 - 5.8 \times 10^{-4}\alpha + 1.0\} \tag{A3}$$

with α in degrees. For the v-component, relation (A2) is confirmed within a few percent when $\|\alpha\| < 45°$.

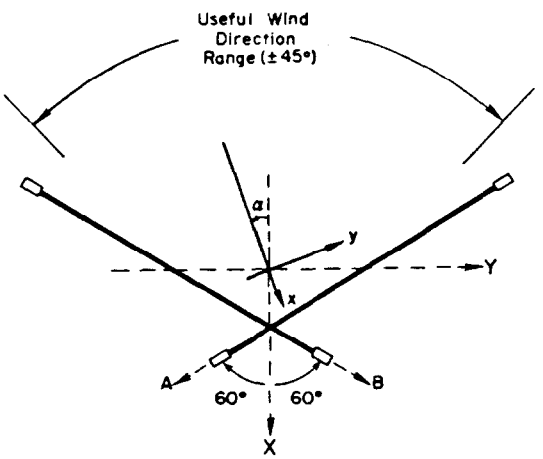


Fig. A1. Relationship between the horizontal axes of a three-axis sonic anemometer and the vector mean axes of the wind (Kaimal, 1979).

The vertical velocity measurement was also calibrated by rotating the sonic anemometer about the horizontal x-axis. Within a few percent, the relation

$$w_m = V \sin \theta \tag{A4}$$

was found to be satisfied, for $\|\theta\| < 45°$, where θ is the elevation angle.

Relations (A2), (A3), and (A4) were used to obtain the true wind vector from u_m, v_m, and w_m. In the field, a measurement was rejected when $\|\alpha\| > 45°$ or $\|\theta\| > 45°$. To minimize the occurrence of rejections, the sonic anemometer was directed into the mean wind so that the average value of α was small.

References

Bruin, H. A. R. de and Holtslag, A. A. M.: 1982, 'A Simple Parametrisation of the Surface Fluxes of Sensible and Latent Heat During Daytime Compared with the Penman-Monteith Concept', *J. Appl. Meteorol.* **21**, 1610–1621.

Buck, A. L.: 1973, 'Development of an Improved Lyman-alpha Hygrometer', *Atmos. Technol.* **2**, 43–46.

Buck, A. L.: 1976, 'The Variable Path Lyman-alpha Hygrometer and its Operating Characteristics', *Bull. Amer. Meteorol. Soc.* **57**, 1113–1118.

Driedonks, A. G. M., van Dop, H., and Kohsiek, W. H.: 1978, 'Meteorological Observations on the 213 m mast at Cabauw, in the Netherlands', *Proceedings on: Fourth Symposium on Meteor. Observ. and Instr.*, Denver, Colo. Amer. Meteorol. Soc., Boston, pp. 41–46.

Friehe, C. A.: 1976, 'Effects of Sound Speed Fluctuations on Sonic Anemometer Measurements', *J. Appl. Meteorol.* **15**, 607–610.

Funk, J. P.: 1959, 'Improved Polythene-Shielded Net Radiometer', *J. Sci. Instr.* **36**, 267–270.

Hanafusa, T., Kobori, Y., and Mitsuta, Y.: 1980, 'Single Head Sonic Anemometer-Thermometer', from: *The Boulder Low Level Intercomparison Experiment*, Preprint of WMO report NOAA/ERL, Boulder, Colo.

Haugen, D. A., Kaimal, J. C., and Bradley, E. F.: 1971, 'An Experimental Study of Reynolds Stress and Heat Flux in the Atmospheric Surface Layer', *Quart. J. R. Meteorol. Soc.* **97**, 168–180.

Kaimal, J. C.: 1969, 'Measurement of Momentum and Heat Flux Variations in the Surface Boundary Layer', *Radio Science* **4**, 1147–1153.

Kaimal, J. C.: 1975, 'Sensors and Techniques for Direct Measurement of Turbulent Fluxes and Profiles in the Atmospheric Surface Layer', *Atmos. Technol.* 7, 7–14.

Kaimal, J. C.: 1979, 'Sonic Anemometer Measurement of Atmospheric Turbulence', *Proc. Dynamic Flow. Conf., 1978*, pp. 551–565, P.O. Box 121, DK 2740 Skovlunde, Denmark.

Kaimal, J. C. and Businger, J. A.: 1963, 'A Continuous-Wave Sonic Anemometer-Thermometer', *J. Appl. Meteorol.* **2**, 156–164.

Kaimal, J. C., Wyngaard, J. C., Izumi, Y., and Coté, O. R.: 1972, 'Spectral Characteristics of Surface Layer Turbulence', *Quart. J. R. Meteorol. Soc.* **98**, 563–589.

Kaimal, J. C., Wyngaard, J. C., Haugen, D. A., Coté, O. R., Izumi, Y., Caughey, S. J., and Readings, C. J.: 1976, 'Turbulence Structure in the Convective Boundary Layer', *J. Atmos. Sci.* **33**, 2152–2169.

Wyngaard, J. C.: 1981, 'Cup, Propeller, Vane and Sonic Anemometers in Turbulence Research', *Ann. Rev. Fluid. Mech.* **13**, 399–423.